# A sparse Gibbs sampler for Optimal Multiple Co-Inertia Analysis: selecting common, partially common and specific dimensions

## 1 Motivation

In OMCOA, each block is approximated as $\mathbf{X}_k \approx \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$, where the diagonal matrix $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \dots, \sigma_k^{(R)})$ measures how much each latent dimension $\mathbf{u}^{(r)}$ contributes to block $k$. The structure of the blocks can be read from the pattern of zeros of $\Sigma_1, \dots, \Sigma_K$: dimension $r$ is

- *common* if $\sigma_k^{(r)} > 0$ for all $k$;
- *partially common* if $\sigma_k^{(r)} > 0$ for some blocks only;
- *specific* to block $k$ if $\sigma_k^{(r)} > 0$ for this block only;
- *inactive* if $\sigma_k^{(r)} = 0$ for all $k$.

OMCOA always returns strictly positive weights, so that this pattern has to be assessed afterwards. We propose a Bayesian version of OMCOA in which a spike-and-slab prior on the weights $\sigma_k^{(r)}$ allows them to be exactly zero. The dimension $R$ is then a *maximal* rank: inactive dimensions are switched off by the sampler, and the effective rank is inferred from the data.

## 2 Model

Let $\mathbf{X}_1, \dots, \mathbf{X}_K$ be column-centered blocks, $\mathbf{X}_k \in \mathcal{M}_{N,p_k}(\mathbb{R})$, and let $R \le \min_k p_k$ be the maximal dimension of the latent space. We denote by $\mathcal{V}_{m,R} = \{\mathbf{W} \in \mathcal{M}_{m,R}(\mathbb{R}) : \mathbf{W}^\intercal\mathbf{W} = \mathbf{I}_R\}$ the Stiefel manifold. The model reads

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad \mathbf{E}_k \text{ with i.i.d. entries } \mathcal{N}(0, \tau_k^{-1}), \qquad k = 1, \dots, K, \tag{1}$$

with $\mathbf{U} \in \mathcal{V}_{N,R}$, $\mathbf{1}_N^\intercal\mathbf{U} = 0$, $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$, and

$$\sigma_k^{(r)} = \gamma_k^{(r)}\, s_k^{(r)}, \qquad \gamma_k^{(r)} \in \{0, 1\}, \quad s_k^{(r)} > 0. \tag{2}$$

**Priors.**

- $\mathbf{U}$ and $\mathbf{V}_1, \dots, \mathbf{V}_K$: uniform on their Stiefel manifolds;
- spike-and-slab on the weights: $\gamma_k^{(r)} \mid \theta_r \sim \text{Bernoulli}(\theta_r)$ and $s_k^{(r)} \sim \mathcal{N}^+(0, \lambda^{-1})$, i.e.

$$p\big(\sigma_k^{(r)} \mid \theta_r\big) = (1 - \theta_r)\,\delta_0\big(\sigma_k^{(r)}\big) + \theta_r\,\mathcal{N}^+\big(\sigma_k^{(r)}; 0, \lambda^{-1}\big);$$

- $\theta_r \sim \text{Beta}(a_\theta, b_\theta)$, one inclusion probability per dimension, shared by the $K$ blocks;

- $\tau_k \sim \mathrm{Gamma}(a_0, b_0)$.

Sharing $\theta_r$ across blocks borrows strength: a dimension that is active in several blocks is more likely to be active in another one, and a dimension that is inactive everywhere sees $\theta_r$ shrink towards zero.

# 3 Full conditional distributions

As in the non-sparse case, the orthonormality of $\mathbf{U}$ and $\mathbf{V}_k$ gives

$$\big\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\big\|^2 = \|\mathbf{X}_k\|^2 - 2\sum_{r=1}^{R}\sigma_k^{(r)}d_k^{(r)} + \sum_{r=1}^{R}\big(\sigma_k^{(r)}\big)^2, \qquad d_k^{(r)} = \mathbf{u}^{(r)\intercal}\mathbf{X}_k\mathbf{v}_k^{(r)}. \tag{3}$$

**Proposition 1** (Full conditionals)**.** *Let $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ be an orthonormal basis of $\mathbf{1}_N^\perp$ and write $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$.*

1. ***Common space:*** $\tilde{\mathbf{U}} \mid \cdot \sim \mathrm{vMF}(\mathbf{Q}^\intercal\mathbf{F}_\mathbf{U})$ *with* $\mathbf{F}_\mathbf{U} = \sum_k \tau_k\mathbf{X}_k\mathbf{V}_k\Sigma_k$.
2. ***Block loadings:*** $\mathbf{V}_k \mid \cdot \sim \mathrm{vMF}(\mathbf{F}_{\mathbf{V}_k})$ *with* $\mathbf{F}_{\mathbf{V}_k} = \tau_k\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$. *When $\sigma_k^{(r)} = 0$, the $r$-th column of $\mathbf{F}_{\mathbf{V}_k}$ is zero and $\mathbf{v}_k^{(r)}$ is uniform on the orthogonal complement of the other columns.*
3. ***Weights (joint update of $\gamma_k^{(r)}$ and $\sigma_k^{(r)}$).*** *Let*

   $$\omega_k = \tau_k + \lambda, \qquad m_k^{(r)} = \frac{\tau_k\, d_k^{(r)}}{\omega_k}.$$

   *Then, with $\sigma_k^{(r)}$ integrated out,*

   $$P\big(\gamma_k^{(r)} = 1 \mid \cdot\big) = \frac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}, \qquad B_k^{(r)} = 2\sqrt{\frac{\lambda}{\omega_k}}\,\exp\Big(\frac{\omega_k\big(m_k^{(r)}\big)^2}{2}\Big)\,\Phi\Big(m_k^{(r)}\sqrt{\omega_k}\Big),$$

   *where $\Phi$ is the standard normal c.d.f.; and*

   $$\sigma_k^{(r)} \mid \gamma_k^{(r)} = 1, \cdot \;\sim\; \mathcal{N}^+\big(m_k^{(r)}, \omega_k^{-1}\big), \qquad \sigma_k^{(r)} \mid \gamma_k^{(r)} = 0, \cdot \;=\; 0.$$

4. ***Inclusion probabilities:*** $\theta_r \mid \cdot \sim \mathrm{Beta}\big(a_\theta + \sum_k \gamma_k^{(r)},\; b_\theta + K - \sum_k \gamma_k^{(r)}\big)$.
5. ***Noise precisions:*** $\tau_k \mid \cdot \sim \mathrm{Gamma}\big(a_0 + (N-1)p_k/2,\; b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$.

*Sketch of proof.* Items 1, 2, 4 and 5 follow as in the non-sparse sampler (linearity of (3) in $\mathbf{U}$ and $\mathbf{V}_k$, and conjugacy). For item 3, the terms of the log-posterior involving $\sigma_k^{(r)}$ are, by (3), $\tau_k d_k^{(r)}\sigma_k^{(r)} - \frac{\tau_k}{2}(\sigma_k^{(r)})^2$. Under $\gamma_k^{(r)} = 0$, $\sigma_k^{(r)} = 0$ and this quantity is 0. Under $\gamma_k^{(r)} = 1$, integrating against the half-normal slab of density $2\sqrt{\lambda/(2\pi)}\exp(-\lambda s^2/2)$ on $(0,\infty)$ gives

$$2\sqrt{\frac{\lambda}{2\pi}}\int_0^\infty \exp\Big(-\frac{\omega_k}{2}s^2 + \tau_k d_k^{(r)} s\Big)ds = 2\sqrt{\frac{\lambda}{\omega_k}}\exp\Big(\frac{\omega_k(m_k^{(r)})^2}{2}\Big)\Phi\big(m_k^{(r)}\sqrt{\omega_k}\big) = B_k^{(r)},$$

which is the Bayes factor of $\gamma_k^{(r)} = 1$ against $\gamma_k^{(r)} = 0$. The conditional of $\sigma_k^{(r)}$ given $\gamma_k^{(r)} = 1$ is obtained by completing the square. □

Updating $(\gamma_k^{(r)}, \sigma_k^{(r)})$ jointly, with $\sigma_k^{(r)}$ integrated out when drawing $\gamma_k^{(r)}$, is essential: drawing $\gamma_k^{(r)}$ given the current $\sigma_k^{(r)}$ would never allow the chain to leave $\sigma_k^{(r)} = 0$. In practice $\log B_k^{(r)}$ is computed on the log scale, with $\log\Phi$ evaluated directly, to avoid overflow.

# 4 Algorithm

The matrix von Mises–Fisher distributions are sampled by one column-wise sweep per iteration, each column being drawn from a vector von Mises–Fisher distribution with Wood's algorithm, as in the non-sparse sampler.

**Algorithm 1** Sparse Gibbs sampler for OMCOA

**Require:** centered blocks $\mathbf{X}_1, \dots, \mathbf{X}_K$; maximal rank $R \le \min_k p_k$; iterations $T$; burn-in $T_0$; hyperparameters $a_0, b_0, \lambda, a_\theta, b_\theta$

**Initialization:** $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ from OMCOA with $R$ dimensions; $\gamma_k^{(r)} \leftarrow 1$; $\theta_r \leftarrow 1/2$; $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
**for** $t = 1, \dots, T$ **do**
  update $\tilde{\mathbf{U}}$ by one sweep of $\mathrm{vMF}\big(\mathbf{Q}^\intercal \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k\big)$; $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  **for** $k = 1, \dots, K$ **do**
    update $\mathbf{V}_k$ by one sweep of $\mathrm{vMF}\big(\tau_k \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k\big)$
    **for** $r = 1, \dots, R$ **do**
      $d_k^{(r)} \leftarrow \mathbf{u}^{(r)\intercal} \mathbf{X}_k \mathbf{v}_k^{(r)}$, $\omega_k \leftarrow \tau_k + \lambda$, $m_k^{(r)} \leftarrow \tau_k d_k^{(r)} / \omega_k$
      draw $\gamma_k^{(r)} \sim \mathrm{Bernoulli}\Big(\dfrac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}\Big)$
      **if** $\gamma_k^{(r)} = 1$ **then** draw $\sigma_k^{(r)} \sim \mathcal{N}^+\big(m_k^{(r)}, \omega_k^{-1}\big)$ **else** $\sigma_k^{(r)} \leftarrow 0$
    **end for**
    draw $\tau_k \sim \mathrm{Gamma}\big(a_0 + (N-1)p_k/2,\ b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  **end for**
  **for** $r = 1, \dots, R$ **do**
    draw $\theta_r \sim \mathrm{Beta}\big(a_\theta + \sum_k \gamma_k^{(r)},\ b_\theta + K - \sum_k \gamma_k^{(r)}\big)$
  **end for**
  **if** $t > T_0$ **then** store $\big(\mathbf{U}, \Sigma_1, \dots, \Sigma_K, \boldsymbol{\gamma}, \boldsymbol{\theta}, \boldsymbol{\tau}\big)$
  **end if**
**end for**
align the stored draws (permutation and signs) on a reference $\mathbf{U}_{\mathrm{ref}}$
**return** the aligned draws

# 5 Posterior summaries

After alignment of the $T - T_0$ stored draws (indexed by $t$) on a reference solution, as in the non-sparse sampler:

- **Posterior inclusion probabilities.** $\hat{P}_k^{(r)} = \frac{1}{T-T_0} \sum_t \gamma_k^{(r),t}$ estimates $P(\gamma_k^{(r)} = 1 \mid \mathbf{X}_1, \dots, \mathbf{X}_K)$. The $K \times R$ matrix $(\hat{P}_k^{(r)})$ summarizes the structure of the blocks; a dimension is declared active in block $k$ when $\hat{P}_k^{(r)} > 1/2$ (median probability model).
- **Effective rank.** $R_{\mathrm{eff}} = \#\{r : \sum_k \gamma_k^{(r)} > 0\}$, whose posterior distribution is estimated by its empirical distribution over the draws.
- **Block weights.** Posterior means and credible intervals of $\sigma_k^{(r)}$; for active entries they are computed conditionally on $\gamma_k^{(r)} = 1$.

## 6 Illustration

We simulated $K = 3$ blocks with $N = 100$, $(p_1, p_2, p_3) = (10, 5, 15)$, 10% of noise and a true rank of 3:

$$\Sigma_1 = \mathrm{diag}(0.7,\, 0.5,\, 0), \qquad \Sigma_2 = \mathrm{diag}(0.6,\, 0.4,\, 0), \qquad \Sigma_3 = \mathrm{diag}(0.5,\, 0,\, 0.6),$$

so that dimension 1 is common, dimension 2 is shared by blocks 1 and 2, and dimension 3 is specific to block 3. The sampler was run with a maximal rank $R = 4$ ($T = 6000$, $T_0 = 2000$, $\lambda = 1$, $a_\theta = b_\theta = 1$). Table 1 reports the posterior inclusion probabilities: the structure is recovered exactly, the fourth dimension is switched off, and $P(R_{\text{eff}} = 3 \mid \mathbf{X}) = 0.96$. All the true non-zero weights lie in their 95% credible intervals.

Table 1: Posterior inclusion probabilities $\hat{P}_k^{(r)}$ (true non-zero entries in bold).

| | Dim 1 | Dim 2 | Dim 3 | Dim 4 |
|---|---|---|---|---|
| Block 1 | **1.00** | **1.00** | 0.02 | 0.01 |
| Block 2 | **1.00** | **1.00** | 0.03 | 0.02 |
| Block 3 | **1.00** | 0.04 | **1.00** | 0.01 |

## 7 Remarks

**Remark 1** (Choice of $\lambda$)**.** *The Bayes factor $B_k^{(r)}$ contains the factor $\sqrt{\lambda/\omega_k}$: a very diffuse slab ($\lambda \to 0$) always favours the spike (Lindley's paradox). The slab precision must therefore be chosen on the scale of the data, e.g. $\lambda = 1$ when the blocks are scaled so that $\|\mathbf{X}_k\|^2$ is of order one, which is the case when each block is divided by its Frobenius norm, as is usual in multiblock analysis.*

**Remark 2** (Choice of $R$)**.** *$R$ must satisfy $R \leq \min_k p_k$, because of the orthonormality of $\mathbf{V}_k$. It should be chosen larger than the expected rank; the superfluous dimensions are switched off. The prior $\mathrm{Beta}(a_\theta, b_\theta)$ controls the expected sparsity; $a_\theta = b_\theta = 1$ is a neutral choice, and $b_\theta > a_\theta$ favours sparser structures.*

**Remark 3** (Mixing)**.** *When $\sigma_k^{(r)} = 0$, the loading $\mathbf{v}_k^{(r)}$ is drawn uniformly, so that $d_k^{(r)}$ is small and the dimension is unlikely to be switched back on in the next iterations. Starting from the full model ($\gamma_k^{(r)} = 1$, OMCOA solution) and letting the sampler switch off the superfluous entries works well in practice; several chains started from different initializations should be compared when the structure is ambiguous.*

**Remark 4** (Link with OMCOA)**.** *When all $\gamma_k^{(r)} = 1$, $\lambda \to 0$ and $\tau_1 = \cdots = \tau_K$, the modes of the full conditionals are the OMCOA updates. The sparse sampler can therefore be seen as OMCOA with an automatic selection of the non-zero entries of $\Sigma_1, \ldots, \Sigma_K$, together with a measure of uncertainty on this selection.*

## References


[1] E. I. George and R. E. McCulloch (1993). Variable selection via Gibbs sampling. *Journal of the American Statistical Association*, 88(423), 881–889.

[2] P. D. Hoff (2007). Model averaging and dimension selection for the singular value decomposition. *Journal of the American Statistical Association*, 102(478), 674–685.

[3] P. D. Hoff (2009). Simulation of the matrix Bingham–von Mises–Fisher distribution, with applications to multivariate and relational data. *Journal of Computational and Graphical Statistics*, 18(2), 438–456.

[4] T. J. Mitchell and J. J. Beauchamp (1988). Bayesian variable selection in linear regression. *Journal of the American Statistical Association*, 83(404), 1023–1032.

[5] A. T. A. Wood (1994). Simulation of the von Mises Fisher distribution. *Communications in Statistics – Simulation and Computation*, 23(1), 157–164.